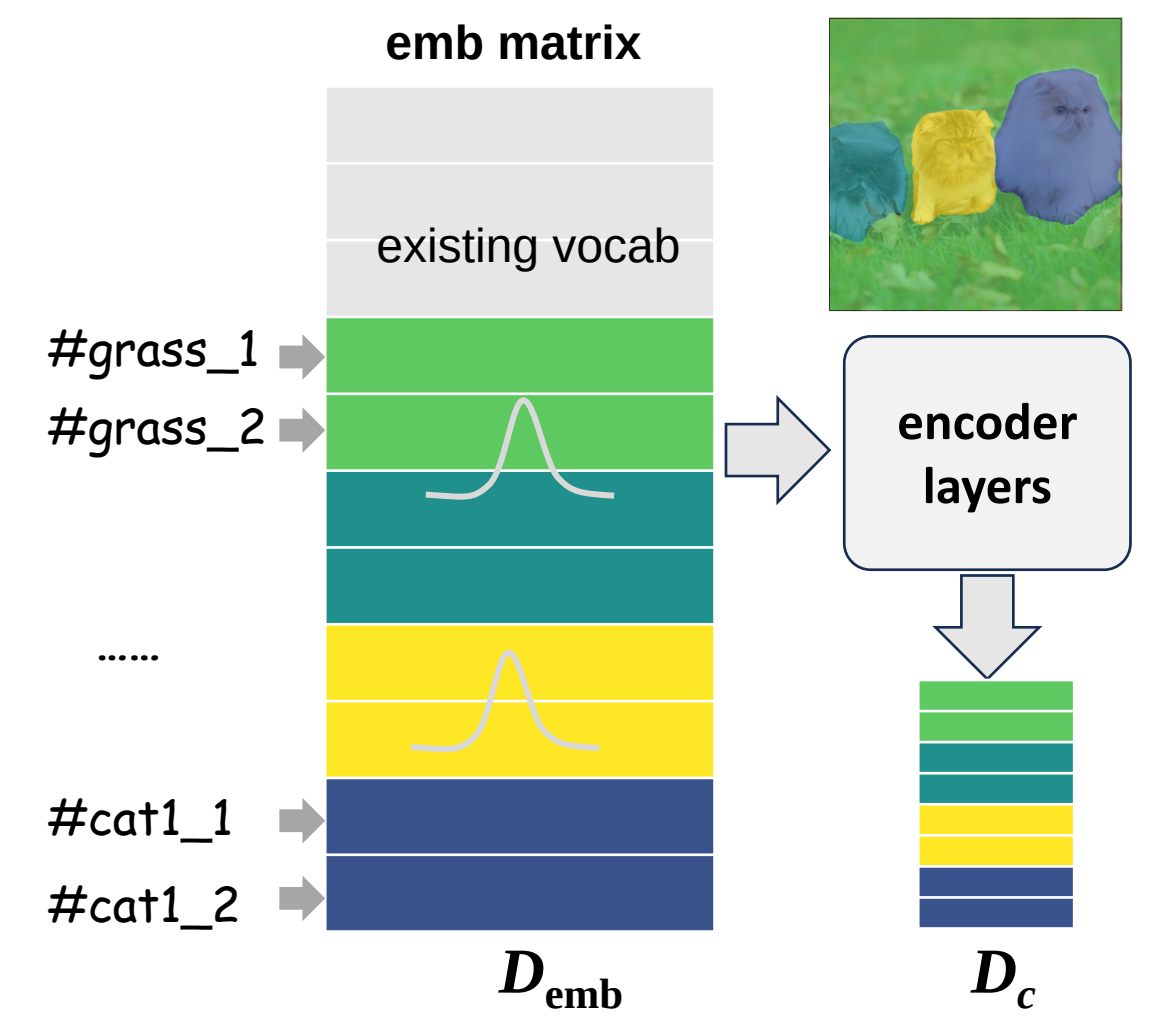
emb matrix
existing vocab
#grass_1
#grass_2
.....
#cat1_1
#cat1_2
encoder layers
D_{emb}
D_c